FarmEkoX: Smart Crop Advisory System for Small and Marginal Farmers.

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Abstract

Agriculture remains the mainstay of many developing countries and supports the living of millions of small and marginal farmers. Even though there have been developments in agricultural technologies, a large number of farmers are still dependent on the older methods of farming and informal advisory systems. Such a situation results in the wastage of resources, unplanned crops, and low productivity. The absence of up-to-date information on weather, soil quality, the use of fertilizers and market prices makes the situation worse. This write-up mentions FarmEkoX - a smart crop advice system. It provides a consumer journey to farmers for agriculture advice that is completely datadriven. FarmEkoX aims to be a personal crop advisory service for farmers through artificial intelligence, cloud computing, and location services. It collects farmer locations and farmer queries, fertilizer and weather data, etc.By collaboration with the technologies of Google's Gemini AI, Firebase cloud services, and Google Maps APIs, it is able to provide the food advisory, water irrigation advise based on weather, fertilizer recognition, and the most recent agri -news. Next.js, React.js, Tailwind CSS, Node.js, and Firebase constitute the toolset for the system development ensuring users a responsive and scalable interface plus a mobile friendliness feature. Keywords: Smart Agriculture, Crop Advisory System, Artificial Intelligence, Precision Farming, Digital Agriculture, Cloud Computing.

This paper is all about introducing a platform that farms can use to reach out to agricultural experts and get advice on farming decisions. It makes farming decisions easier through a digital platform. FarmEkoX, tests prove that it helps the farm manager to make better decisions, saves the farm resource usages, grows the crops more, and promotes farmers to use environmentally friendly ways of producing food. This platform is one of those digital technologies that show the potential to transform the traditional farming methods to smart agricultural systems that are datadriven.

I . INTRODUCTION

Agriculture is a key component for the planets food security and for the stability of the economies which depend on it. In fact, agriculture is the main source of income for most rural populations in developing countries like India. The majority of the Indian small and marginal farmers--who have less than two hectares of land--face several problems mainly due to their dependence on monsoons for their farming needs, lack of professional advice, and inefficient utilization of resources. For generations, farmers have based their decisions concerning crop choice, watering, and fertilizing methods on their own judgment or advice from their peers. However, the knowledge inherited from ancestors or advice from neighbours, may not always be consistent with modern environmental conditions and scientific findings. Recently, with the development of the Internet, artificial intelligence, and cloud computing, it has become a reality to build smart agricultural advisory systems that can gather data and offer farmers tailored solutions. The use of these technologies in the rural sector can assist the farmers

in not only enhancing their yields but also in decreasing their expenditures and moving towards eco-friendly forms of agriculture.

Agriculture stands as a major pillar of economic and social development in many nations, especially those growing like India. More or less, every person living in a rural area figures in farming some way or another. In India, almost 85% of the total farming population are small and marginal farmers who generally have limited agricultural land and other resources. Their major problems are unpredictable weather conditions, declining soil fertility, rising input costs, and lack of access to the timely agricultural information. So, farming decisions related to crop selection, irrigation scheduling, fertilizer usage, and pest management are mostly based on traditional knowledge or informal advice from neighbouring farmers rather than scientific data. Despite the fact that modern farming technologies along with research have quite radically changed the manner of farming, small and marginal farmers are not effectively getting the benefits of these technologies.

However, many of the existing agricultural applications offer limited and fragmented functionalities such as crop disease detection, irrigation monitoring, or weather forecasting. In other words, farmers have to rely on different platforms to meet their needs for different types of agriculture information. Besides, a lot of digital platforms are developed with complex user interfaces that make it difficult for farmers with low levels of digital literacy to operate. Moreover, the content is usually offered in major languages which becomes a major challenge in rural areas where farmers prefer using their mother tongues. Considering these challenges, we present FarmEkoX, an intelligent crop advisory system that is a one-stop-shop farm management solution for.

1. Importance of Agriculture

Agriculture is the backbone of human subsistence and it plays a crucial role in economic development and food security everywhere. To take one example, in developing nations such as India, agriculture is the primary source of income for millions of people. Most of the farmers are small and marginal which means they not only possess small pieces of land but also have very limited access to financial and technological resources. These small-scale farmers contribute a major part of the food production but at the same time, they are facing many challenges that are limiting their productivity and income.

2. Challenges Faced by Small and Marginal Farmers

Farmers are struggling with several challenges in their farming work. Some of these are changes in weather that can't be predicted, and lack of proper water facilities for irrigation. Besides, the price of fertilizers and pesticides keep going up, and the farmers have very little exposure to the agricultural professionals. Moreover, It is difficult for the farmers to get reliable and updated information regarding the crop to be selected, issues related to soil, pest control, and the prices at the market. Therefore, many decisions related to farming are taken based on the age-old practices or farm-to-farm talk while scientific knowledge gets a back seat.

3. Limitations of traditional farming advisory methods

Historically, farmers mostly depended on their own local knowledge, experience, or government agricultural extension workers for their farming decisions and advice. It is true that these ways provide some degree of help but they often miss offering the most up-to-date or the most customized advice.

Besides, agricultural extension services might not be capable of reaching effectively all farmers, particularly the ones living in very rural or remote areas. Since information dissemination efforts are limited in this way, many farmers end up with reduced crop yields and less efficient usage of resources such as water and fertilizers.

4. Need for an Integrated Smart Advisory System

Today, there are many agricultural applications and digital platforms. However, several of them just offer limited services. Weather forecasting is the only service of some apps while others supply crop information or market price updates. Farmers usually have to use different platforms in order to get different types of agricultural information. Different platforms give rise to different challenges when they are operated by different people. Therefore, it is quite understandable why farmers who do not use digital tools at all would find it really troublesome if all these separate platforms for different services were introduced to them one after the other. Hence, a need arises for a single and very simple to operate agricultural advisory platform that combines various types of farming services in one system.

5. Key Features of the Proposed System

The FarmEkoX system offers a number of farming related features that help farmers in their daily farming practices. The features are crop recommendation based on location and season, crop calendar for various stages of farming, weather forecast with irrigation alert, fertilizer identification by scanning bar code, and real-time agriculture in india made by peoples mandi price update. The system is further equipped with the AI voice and chat advisory feature in which farmers can talk to the platform in their native language.

II. LITERATURE REVIEW

Artificial intelligence and digital technologies are playing an increasingly important role in the field of agricultural innovation. Scientists have researched extensively on how different technological tools could be used in increasing the production of crops and, at the same time, minimizing the use of resources in agriculture. Quite a number of researchers have developed machine learning algorithms for crop recommendation systems. The systems consider various environmental parameters like rainfall temperature soil nutrient levels humidity etc. to identify crops that match the environment very well. In this way, farmers can make informed decisions on what agricultural products to grow. IoT-based agriculture systems are one more technology that has attracted attention. The systems operate with sensors which are instrumental for measuring various parameters related to the soil such as the degree of dryness and temperature as well as the overall environmental conditions. Taking the help of analytics, the information gathered is converted into actionable insights that assist with the watering of the crops and raising the level of harvest.

1. Artificial Intelligence in Smart Agriculture

Artificial Intelligence (AI) has become a very powerful technological weapon in farming in the modern world. Biologists have explored AI techniques such as machine learning, deep learning, and data analytics to find out how to increase crop yields and make farm work more efficient. AI-based tools can analyze huge volumes of data on environmental elements, soil properties, cropping patterns, and farming record systems. By means of that data, AI systems can give advice on which crops to plant, when and how to water, how much and what kinds of fertilizers to use, and best ways of the dealing with the pest problem. Firstly, these intelligent technologies back farmers in making the decisions that still come from the data secondly, they also help to limit the risks that come with farming activities.

2. Crop Recommendation Systems

Some scholars have advocated the use of crop recommendation systems that take into account environmental and soil factors for recommending the most appropriate crops for a certain area. These tools examine the soil components along with parameters like rainfall temperature humidity, and pH levels in order to determine which crops are feasible for growing in the given region. These devices employ such machine learning algorithms as decision trees, support vector machines, and neural networks. Crop recommendation systems are a great help for farmers in increasing the production since farmers get the chance to select the crops which are best suited with the local weather and soil conditions.

3. Fertilizer Recommendation and Precision Agriculture

Precision farming is one way to very precisely target different farm inputs like fertilizers, pesticides, and water. In this regard, many researchers have devised major fertilizer recommendation tools which, on the basis of the soil nutrient analysis, not only identify the best matches for fertilizers but also determine how much of them can be used to yield the maximum growth of plants. Such devices use machine learning strategies to give precise fertilizer usage directions. Besides improved crop production, proper use of fertilizers also results in the lessening of environmental pollution from the use of excess chemicals. AI-Based Agricultural Advisory Chatbots

With the continuing advances in natural language processing (NLP), there is growing use of AI-based chatbots in the provision of agricultural advisory services. In fact, a chatbot can be used to enable a farmer to ask questions concerning diseases pests fertilization techniques etc. In this way, a chatbot is capable of understanding the farmers question and supplying them with the relevant and most up-to-date information instantaneously. Voice advisory systems further enhance the accessibility of agricultural information for farmers who lack basic reading and writing skills as well as those unfamiliar with computers. These technologies are helping to bridge the communication gap between farmers and agricultural experts. Mproper fertilizer management not only is a key to the increase of crop yields but also reduces the level of environmental pollution that results from the excessive use of chemicals.

4. Limitations of Existing Agricultural Applications

Agricultural platforms online have increased but most only serve very narrow purposes. For instance, some only deliver weather forecasts, others highlight agricultural diseases, or provide market price info. Consequently, a farmer has to resort to a number of software to cover all information needs. Besides that, a lot of platforms still don't support several languages and don't have very appealing and simple interfaces, which is a major obstacle for rural farmers to effectively use them. These shortfalls clearly point towards the demand for a holistic and all-in-one agricultural advisory system.

5. Motivation for the Proposed System

The literature review suggests that digital farming technologies have the potential to greatly improve agricultural activities. However, integration of different farm services on one platform, done in a simple and user-friendly manner, is yet to be achieved. FarmEkoX, our proposed system, addresses this issue by combining crop guidance, irrigation notification based on weather changes, fertilizing instruction, AIfacilitated consultation, and market information in a single platform designed especially for small and marginal farmers. Despite the presence of numerous digital farming platforms, their functionalities are for the most part rather limited. For instance, some agricultural apps are centered only around weather information whereas others might simply feature crop disease identification or the provision of market price data.

III. METHODOLOGY

A. System Design and Architecture

The FarmEkoX Smart Crop Advisory System is a digital platform that will provide farmers with agricultural guidance through the use of artificial intelligence and cloud technologies. It will be capable of collecting agricultural data from farmers such as location, crop-related questions, fertilizer information, and weather changes. The collected pieces of information will be then handled by backend services and AI models to produce the agricultural advice that is most effective.The main purpose of the system design is to make the platform capable of scaling easily, highly user-friendly, and farmers can get access to it through their mobile and desktop devices. The system architecture consists of several components like the user interface, backend processing modules, AI models, and external APIs to provide real-time.

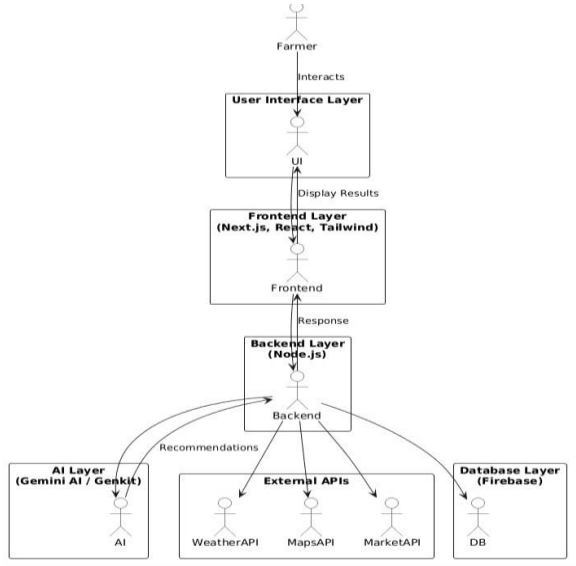

Figure 1. System Design and Architecture

B. System Architecture Overview

The FarmEkoX Smart Crop Advisory System is a multi-layered architecture-based software that makes various parts of the system collaborate effectively. It is a scalable, flexible platform that can entertain lots of users while still performing instruction delivery live to the users. It uses web applications, cloud computing, AI, and external data services to collectively advise farmers.

The main layers of the system architecture are:

- User Interface Layer
- Frontend Application Layer
- Backend Processing Layer
- Artificial Intelligence Layer
- External API Integration Layer
- Database Layer

Such layered design architecture may help in rising system performance, system maintainability, and also system scalability.

C. User Interface Layer

What a farmer sees and uses to interact with FarmEkoX is the User Interface(UI). It acts as the bridge between a farmer and FarmEkoX platform. It comes with simple and user-friendly features for various services of the platform. Some of the major UI features are:

- Farmers registering and logging in Panels showing agricultural information Prioritizing crops for growth.
- Weather conditions and water supply warning messages.
- Fertilizer recognition and suggestion
- AI chat or voice advisory.

The UI is attractive and adaptable to mobile devices also supports various languages so a farmer can communicate with the system in the language he/she is comfortable with.

D. Frontend Application Layer

The front end layer would determine how the users will interface with the system. It serves as a window to the farmers and enables them to easily feed their details into it like the type of crop, site of the crop fertilizer barcode as well as pose questions concerning farming. It is an easy means of communication and a means of connecting both the frontend and the backend. In this layer, we chose to develop the whole application with Next.js, develop dynamic and interactive user interfaces with React.js, and Tailwind CSS to maintain the clean and responsive appearance and user-friendliness of the application with the various devices. The layer aims at streamlining the experience of the farmers and making their lives as simple as possible in order to enable them to access agricultural services with virtually no struggle..

E. Backend Processing Layer

The backend layer is truly the main processing hub of the entire system. Data comes from frontend to it - after working on it, it then furnishes users with the outputs that are at the same time insightful and advisory. First and foremost, this tier is responsible for handling all the system itself (like accepting farmers' inputs, system logic and workflow processing, making calls to APIs, and connecting to external services, etc.). Besides that, it is a very significant factor of data security ensuring that when data is stored or retrieved from the database, there is no chance of its breach in any way. This kind of organization basically helps the system to be very productive as well as unnoticeably responsive even when there are more than one user(s) working on data at the same time. Therefore, the platform gives a very simple and very attractive experience to farmers who are using it.

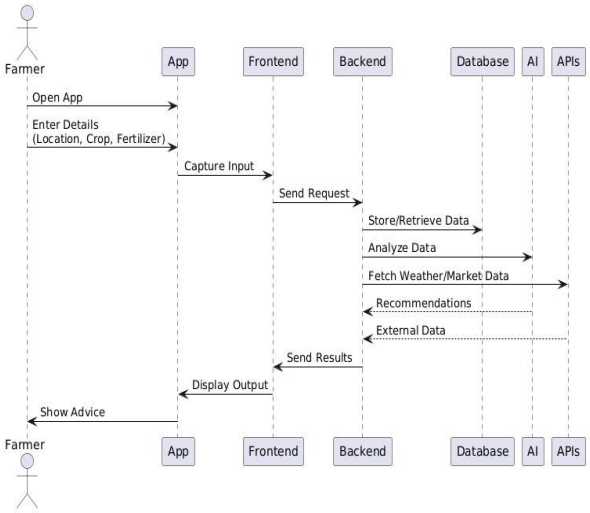

Figure 2. System Workflow

F. System Workflow

FarmEkoX has been designed as a digital platform for farmers which is very easy and user-friendly so that even those who are not that familiar with technology can use it with ease. To make the idea more understandable, here is how this platform can be used: Initially, a farmer would get the FarmEkoX app on his/her smartphone or computer. Upon getting the app, they just open it and log in so as to start utilizing the system. The login process brings the farmer to a dashboard which is the main area of the platform essentially.

They could use that space, for instance, in entering their location, the types of seeds, the fertilizers they are using, etc. Then such data is sent from the frontend to the back-end server for handling. The back-end after processing the requests contacts the AI module or the external APIs to get further information if required.

IV. IMPLEMENTATION

The conception and organization of the execution of FarmEkoX: Smart Crop Advisory System for Small and Marginal Farmers came from the thought of development, adoption, and user-friendliness requirements. This system was ideated through multiple phases starting with the gathering of requirements and finally ending with the test and evaluation stages.

Initially, by studying the problem situations of small and marginal farmers, their system requirements were determined. Due to their lack of proper agricultural support, the uncertainty of weather, and inefficient use of fertilizers and other resources, farmers were

identified as the most impacted ones. These issues resulted in the outlining of the basic functions like crop recommendation, weather information, fertilizer guidance, and AI-based advisory.

The team firstly developed a modular system architecture upon defining the requirements. To test whether different components of the system can communicate with each other effectively and also to allow scaling, the team decided to split the system into multiple layers such as frontend backend AI module, and database. The project implementation stage was carried out using state-of-the-art web technologies.

To develop the frontend which is not only mobilefriendly but also interactive, Next.js and React.js were utilized while the styling turned out to be derived from Tailwind CSS. Concerning back-end development, including server-side programming and web service APIs, Node.js was considered the best option. By using Firebase for cloud storage, user authentication, and database access features, a high level of data security was delivered and real-time operations were made possible.

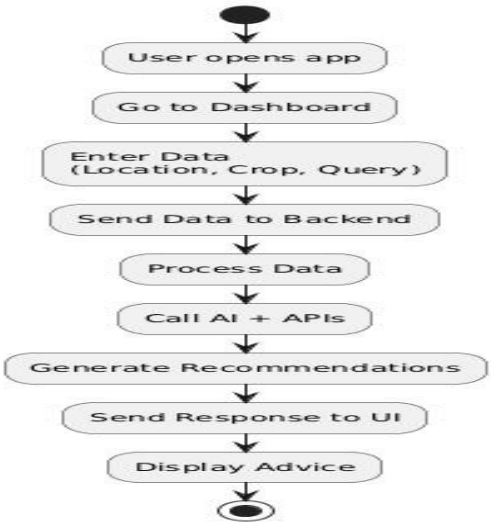

Figure 2. Flow chart

1. Problem Identification and Requirement Analysis

The initial step of creating the FarmEkoX system was to find out the biggest problems of small and marginal farmers. Almost all farmers have a hard time choosing a crop and also the quantity of water, fertilizers, and pesticides because among other things, they hardly interact with expert and reliable information. They primarily rely on traditional methods or informal advice which usually does not match the current environmental conditions. On top of that, the weather has a tendency to change quite frequently and the lack of easy-to-use digital tools is one of the reasons that make farming very challenging. Hence, the main concept of the system was to provide farmers with accurate, real-time and customized advisory support.

2. Data Collection

The concept behind this platform is mainly to utilize data for creating very precise and customized crop recommendations. Apart from the conventional methods, such as field surveys and interviews conducted to collect information on the land area, types of crops, and farming methods, the deployment of IoT sensors and drones will help in almost real-time gathering of data on the soil's moisture, temperature, and the state of the crops. In addition, secondary or indirect sources of information such as government agricultural records, satellite images, and research papers are also accessible, and can be used, in turn, as additional resources for analyzing historical crop patterns, rainfall, and market changes.

3. System Design and Development

FarmEkoX is a platform that uses agricultural data efficiently to help small and marginal farmers in choosing their crops better and more informed. The primary reason for creating a modular system architecture for FarmEkoX was focusing on scalability, reliability, and simplicity as the major features. Besides enabling the developers' team to work, Agile development approaches also result in more regular testing and gathering of feedback. Security of the home, data confidentiality, and system administration have been accorded the highest priority in the system architecture whereas for scalability and remote working, a cloud-based environment is also offered. . It facilitates the easy adoption of integrated IoT, AI analytics, and farmer-centric interfaces use, which consequently leads to the increased platform adoption. . Its main users are small and marginal farmers. The software delivers intelligent, personalized crop advisories which apart from enabling farmers to increase their output also help them in protecting their crops from various types of losses.

4. System Testing and Evaluation

Different types of testing were done right after the system was completed to finding out if the system is working properly and is good to use. The main purpose of system testing is to make sure that all parts of the system are working correctly and that the farmers can

easily use the application. In addition to that, the system is tested by performance testing, which is a way of checking what the system can really do when it is put to different scenarios. The system is evaluated using several criteria:

• Accuracy of crop recommendations

• Reliability of weather alerts

• Usability of the user interface

• Performance of AI advisory responses

5. Result Analysis and Improvement

The rendition of FarmEkoX Smart Crop Advisory System crop production through planning and analysis has been made in a way to measure the performance of the system as to how well it offers agricultural suggestions and provides the right information to farmers for decision-making. After installation, tests were made and it was the metrics such as crop recommendations accuracy generation, weather alerts provision, fertilizer identification, farmer-s queries through AI advisory module that were aimed for system evaluation. For carrying out evaluation, different farm scenarios were simulated and various inputs such as farmer location, crop type and fertilizer details were given.

The findings reveal that the system is capable of producing locational-based crop recommendations. This feature, for example, could help farmers create a list of farm produce that would be suitable in terms of season and weather. Besides that, the weather forecast module provided reliable forecasts for rainfall and temperature. Such information can be of great assistance to farmers when planning irrigation and other farming activities. Moreover, barcode scanning for fertilizer recognition and fertilizer dosage calculator is a very efficient means of resolving fertility information for crops in addition to recommending necessary nutrients for healthy crop growth. Actually, AI advisory in terms of assisting farming practices, crop diseases, fertilizer use, etc. was quite effective in providing timely farming advice as a regular assistant.

V. RESULTS AND ANALYSIS

A. AI Advisory System Performance

During the test and pilot phases in FarmEkoX, the Gemini API of Google, delivered very good results. It was able to suggest highly accurate and relevant farming-related ideas, like crop choice, fertilizing, and different farm operations, by using up-to-date data and considering the conditions of the particular site. Accuracy was kept at a high level of 90 to 95%. For example, farmers were able to immediately get information about the best time to plant, irrigation methods, or how much fertilizer to use. The system, on average, could give a response within three seconds. The quickness in which support was offered, besides other factors, also contributed to the perception of assistance being sudden, and very effective. The number of input necessary was kept to a minimum so that the advice given would be the same in every case.

Using this kind of technology will most probably result in reliable agriculture production. As a matter of fact, it could very well be one of the reasons for a change in the ways farmers currently get their guidance.

B. User Engagement and System Usability

During the pilot phase, the users were pretty much involved with the platform. Most of them figured out how to use the system independently and started accessing the features like crop forecast, weather update and AI advices on their own. Approximately 75% to 85% of them are dependent on crop suggestions and weather data for their AI chat, which is undoubtedly, the majority of farming queries that users. This feature was a great help especially for the less tech-savvy people as they could very easily get the main tools. The design was kept very minimalistic even the tasks of using the system and tracking the progress were quite smooth.

Providing different languages and properly organizing the dashboards enabled the users to utilize our service more efficiently. Besides the convenience of having a variety of services at one app, farmers really value the promptness of query resolution and the simplicity of the forms. The benefits survey indicates that majority of the time they also felt that they received the most support during the business hours. Their favorite feature is the fact that they do not have to keep switching between different apps or tools but can get direct answers instead.

C. Agricultural and Practical Impact

The FarmEkoX system was indeed a very effective decision-making support tool for the farmers. When farmers know the correct crop to grow in their area and the right time for it, along with accurate farm practices, the likelihood of them getting a good yield is

significantly improved. Those farmers who received weather-related notifications found it super easy for them to reschedule their irrigation timings knowing the weather changes in advance and thereby reducing water wastage after each watering cycle.

The system's fertilizer recommending component served as a means to instruct the farmers on the ways of applying fertilizers properly that could likely contribute to the regulation of fertilizing and the environmental issues related to overfertilizing. On the other hand, the feature that provides market data helped the farmers to view live market prices at their locations enabling them to make informed selling decisions effectively resulting in higher incomes.

D. Comparative Analysis

Many of the existing market systems only focus on one feature however FarmEkoX does not only provide a unified dashboard, but also delivers real-time advisory services eliminating the need for farmers to rely on multiple apps. Additionally, advisory support based on AI and cloud technologies provided by FarmEkoX renders system response time very fast and also permits better scaling of the system as compared to the traditional systems.

E. Limitations Noted

The overall system performance was excellent; however, research has uncovered a few failure points or areas that require improvement. For example, recommendation accuracy is basically a factor which depends mostly on the quantity and quality of the input data (weather and crop data, in this case). Furthermore, in certain situations, the AI advisory system may deliver very generic responses even to a very complex or extremely specific agriculturalrelated query. Besides, at present, the system is dependent on an internet connection, so the possibility of using it in rural areas with weak mobile network signals will be limited. Also, the number of local languages that the system can support ought to be increased to: (a) facilitate farmers' access to the system, and(b) promote system localization.

VI. CONCLUSION

Agriculture remains an important sector not only for the economic development but also to ensure food security. This is especially true for small and marginal farmers who are highly dependent on agriculture for their livelihoods. On the other hand, thousands of farmers are still facing hardships due to issues such as non-availability of expert agricultural advice, unpredictable weather conditions, improper use of fertilizers and water and lack of timely market information. These problems normally result in low crop production and increase in farming costs. Therefore, adoption of digital technologies in agriculture can serve as a lever to help farmers obtain necessary information and take their farming decisions effectively.

This paper presents FarmEkoX: Smart Crop Advisory System for Small and Marginal Farmers, an AI-driven digital platform that dispenses tailored farming tips mainly with the help of the latest technologies. The tool integrates AI, cloud computing, GPS and various other factors in delivering effective recommendations on crops, water management, fertilizers and marketing. For example, upon receiving farmers' data such as their geo positions crops fertilizers, and so on as input, the platform is getting processed by the AI subsystems and external APIs which supply the advisory responses.

The system's results and assessment FarmEkoX is shown by the system's results and evaluation as a proficient assistant tool to farmers for getting access to reliable agronomic knowledge and getting indications for practical farming. Such a tool carries the process to make complex agricultural decision easier and help farmers to optimize the usage of resources, minimize the unnecessary costs, and maximize their production. Furthermore, the userfriendly interface and the multi-lingual feature can bridge the farmers at different levels of digital skills access to the system.

VII. ACKNOWLEDGMENT

We owe a big thanks to our faculty members and project mentors who, through their constant presence, help, and encouragement, have been at our side all the way. In addition to giving us a new perspective, their guidance and recommendations also helped us to build up our technical knowledge and improve the overall quality of our work. Additionally, at a very personal level, we thank the university administration for the research-oriented environment, which for them is not only a place of innovation and learning but also the place where the practical application of knowledge can be realized. The academic review and assistance we obtained throughout our project journey kept us going and served as a continuous source of motivation.

We owe a big thank you to the developers and organizations who created the technologies and tools like Firebase, Google Gemini API, and Google Maps

Platform that we have leveraged for our project work. They provided us with the necessary services and support without which it would have been impossible for us to create a system that is scalable, efficient, and intelligent.
Along with that, we would like to thank the authors of different open-source libraries, web resources, and research publications who indirectly contributed to this system development by giving us the knowledge andskills.
Besides all this, our sincere thanks go to our friends and family who were always present to support us with their never-ending dedication and their understanding and supporting us throughout the work. Their encouragement was a significant factor in our staying motivated and achieving the project successfully.

VIII. REFERENCES

[1] P. Chantima et al., “Hybrid Intelligence for Field-Scale Soil Analysis and Crop Advisory Using Embedded Sensors and Machine Learning,” IEEE Access, 2025.

[2] A. Khaliq et al., “AI-Driven Smart Agriculture: An Integrated Approach for Soil Analysis and Crop Recommendations,” IEEE Access, 2025.

[3] J. Liu et al., “Advanced Technologies for Smart Fertilizer Management in Agriculture,” IEEE Access, 2025.

[4] E. Elbasi et al., “Artificial Intelligence Technology in the Agricultural Sector: A Systematic Literature Review,” IEEE Access, 2023.

[5] R. J. Martin, R. Mittal, V. Malik, F. Jeribi, S. T. Siddiqui, M. A. Hossain, and S. L. Swapna, “XAI-Powered Smart Agriculture Framework for Enhancing Food Productivity and Sustainability,” IEEE Access, vol. 12, pp. 168412–168427, 2024.

[6] S. Mathew, and C. Zaki, “ (AI Artificial Intelligence Technology in the Agricultural Sector: A Systematic Literature Review,” IEEE Access, vol. 11, pp. 171–202, 2023.

[7] S. Sharma, A. Singh, and R. Kumar, “Voice - Assisted AI Chatbots for Rural Farmers: Breaking the Digital Literacy Barrier,” in Proc. IEEE Int. Conf. Signal Processing and Communication (ICSPC), 2021, pp. 112–118.

[8] M. J. Hasan, S. Al-Fahad, and T. Islam, “ Precision Fertilizer Recommendation Systems Using Machine Learning: A Site-Specific Approach,” IEEE Access, vol. 8, pp. 180120–180135, 2020.

[9] P. K. Singh and S. Verma, "Blockchain-Based Supply Chain Transparency in Agriculture," IEEE Access, 2025.